# Nutrition & Fluid Therapy (Surgical)

## The Big Picture

Surgical patients starve, get stressed, and lose fluid — and **malnutrition (30% of GI surgical patients, 60% if complications)** worsens healing and outcomes. Know the **metabolic response to starvation vs trauma**, daily **fluid/electrolyte** needs, when to feed **enterally vs parenterally**, short bowel syndrome, and **refeeding syndrome**.

## Metabolic Response to Starvation

- **First 12 h** — liver glycogenolysis; brain/RBC/WBC/renal medulla use only glucose; Cori cycle (lactate→glucose).
- **>24 h** — glycogen depleted → **gluconeogenesis** from glutamine/alanine (up to 75 g muscle/day).
- **Prolonged** — fat oxidation, glycerol→glucose, hepatic ketones.
- **48–72 h** — CNS switches to **ketones** ("fat fuel economy"), muscle breakdown falls to ~55 g/day; **REE drops** (↓ T4→T3 conversion).
- **Trauma/sepsis** — different: impaired ketogenesis, protein breakdown for glucose, **cannot be fully prevented by giving glucose**.

## Nutritional Assessment = ABCD

- **A**nthropometry — BMI, mid-arm circumference, skinfold.
- **B**iochemistry — albumin, CRP, WBC; Hb (anaemia); Ca/phosphate (refeeding risk).
- **C**linical — GI symptoms, comorbidities (cancer, IBD, liver, stroke).
- **D**ietary — 24-h recall; **MUST** (Malnutrition Universal Screening Tool).

## Fluids & Electrolytes (healthy 70 kg/day)

- **Water 2.2 L**, **Na■ 0.9–1.2 mmol/kg**, **K■ 1 mmol/kg**, Ca 5 mmol, Mg 1 mmol.
- Losses ↑ with pyrexia, tracheostomy (insensible), diarrhoea, fistula.
- IV fluids: **0.9% saline, Hartmann's, 5% dextrose** ± colloids — none perfectly mirror plasma. Assess with pulse, BP, CVP, I/O charts, electrolytes.

## Nutritional Requirements

- **Energy 20–30 kcal/kg** (~1300–1800 kcal/day in hospitalised).
- Carbs 45–65% of calories; fat = LCT emulsions.
- **Protein/nitrogen 0.10–0.15 g/kg** (0.20–0.25 if hypermetabolic).
- Vitamin C 60–80 mg; **B12 after intestinal/gastric resection**; A,D,E,K reduced in steatorrhoea.

## Intestinal Resection

- **Up to 50% of small bowel** can be removed without permanent harm.
- **Ileum** = only site of **B12 + bile salt** absorption → resection → bile salt loss → fat malabsorption + diarrhoea (needs parenteral B12).
- Proximal jejunum resection → ileum/colon adapt (minimal effect).
- **Short bowel syndrome** — >200 cm resected + colectomy, OR <150 cm remaining. **Net absorbers** (>100 cm jejunum, manage without parenteral) vs **net secretors** (<100 cm, output >4 L/day, need supplements).

- SBS complications: peptic ulcer (gastric hypersecretion), **gallstones** (bile salt loss), **hyperoxaluria** → **renal stones**. Treat: low-carb diet, **anti-secretory (PPI/octreotide), anti-motility (loperamide/codeine)**.

## Artificial Nutrition

- **Enteral first** ("if the gut works, use it") — preserves mucosal barrier, prevents atrophy, fewer infections, shorter stay.
- Routes: **sip feeds**, NG/fine-bore tube, **PEG** (if >4–6 weeks needed), nasojejunal/jejunostomy.
- Formulas: polymeric (intact protein), monomeric/elemental (amino acids), immunonutrition (glutamine, arginine, fish oil).
- Complications: tube (malposition, blockage), GI (diarrhoea, aspiration), metabolic.
- **Parenteral (TPN)** — when gut can't absorb (short bowel, fistula). 3-L bag (lipid + amino acids + glucose + electrolytes + vitamins). Give via **central line (SVC/RA)** — femoral avoided (infection). CXR confirms tip. Complications: insertion (pneumothorax), line **(sepsis, thrombosis)**, metabolic (**refeeding**, liver dysfunction, bone disease).

## Refeeding Syndrome

- **Risk:** BMI <18.5, >10% weight loss in 3–6 mo, little intake >5 days, alcohol/insulin/chemo/diuretics.
- **Mechanism: hypophosphataemia** + fluid/electrolyte shifts → arrhythmias, muscle weakness, cardiac/respiratory failure, seizures; can be fatal.
- **Prevention:** correct electrolytes first, start feeding **low and slow**, monitor electrolytes.

## Walk-Away Points

- Starvation → ketone economy (saves muscle); trauma → obligatory protein catabolism.
- **Ileum = B12 + bile salts**; lose it → diarrhoea, gallstones, renal stones.
- **Enteral beats parenteral** whenever gut works; TPN via central line.
- **Refeeding syndrome = hypophosphataemia** → feed low and slow in at-risk patients.

## Nutrition & Fluid Therapy — Revision Table

| Starvation phase | Fuel |
|---|---|
| First 12 h | Liver **glycogenolysis** + Cori cycle |
| >24 h | **Gluconeogenesis** (glutamine/alanine), 75g muscle/day |
| 48–72 h | **Ketones** (CNS), muscle loss ↓ to 55g/day, REE ↓ |
| Trauma/sepsis | Protein catabolism, **not fixed by glucose** |

| Assessment (ABCD) | Marker |
|---|---|
| Anthropometry | BMI, mid-arm circ, skinfold |
| Biochemistry | Albumin, CRP, Ca/phosphate |
| Clinical | GI symptoms, comorbidities |
| Dietary | 24-h recall, **MUST** |

| Daily need (70kg) | Amount |
|---|---|
| Water | **2.2 L** |
| Sodium | 0.9–1.2 mmol/kg |
| Potassium | 1 mmol/kg |
| Energy | **20–30 kcal/kg** |
| Protein/N | 0.10–0.15 g/kg (0.20–0.25 stressed) |

| IV fluids | 0.9% saline, Hartmann's, 5% dextrose, colloids |
|---|---|

| Intestinal resection | Consequence |
|---|---|
| **Ileum** | Only site **B12 + bile salts** → diarrhoea, fat malabsorption |
| Jejunum (proximal) | Ileum/colon adapt — minimal effect |
| Short bowel | >200cm + colectomy OR <150cm left |
| Net absorber | >100cm jejunum, no parenteral |
| Net secretor | <100cm, output >4L/day, supplements |
| SBS complications | Gallstones, hyperoxaluria→renal stones, peptic ulcer |

| Enteral nutrition | Detail |
|---|---|
| Rule | **Gut works → use it** |
| Routes | Sip, NG/fine-bore, **PEG (>4–6 wk)**, jejunostomy |
| Benefits | Mucosal barrier, fewer infections, shorter stay |

| Parenteral (TPN) | Detail |
|---|---|
| Indication | Gut can't absorb (short bowel, fistula) |
| Access | **Central line (SVC/RA)** — not femoral |
| Complications | Sepsis, thrombosis, **refeeding**, liver dysfunction |

| Refeeding syndrome | Detail |
|---|---|
| Mechanism | **Hypophosphataemia** + electrolyte shifts |
| Risk | BMI<18.5, >10% loss, >5d no intake, alcohol |
| Prevention | Correct electrolytes, feed **low & slow** |

***Bold = high-yield.***